

Malocclusion and Jaw Joint Disorders: A Look into Auto-segmentation and Quantification of Medical Scans





01.

Motivation

Current Conditions^[1][9]

Human Factors

- Tedious
- Labor Intensive
- Lost Opportunity Costs
- Prone to Errors/Speciality
- Human Resource Costs
 - Wage
 - Training
 - Labor Quality
 - Location

Time Factors

- Slow
- Ineffective
- Wasted Time Gaps
- Inconsistent Production Rate
- Delays



03.

Teeth Segmentation

How can we find these?

- Each person has a different jaw-shape.
- Each person has a different tooth shape.
- Each person has a different teeth alignment.

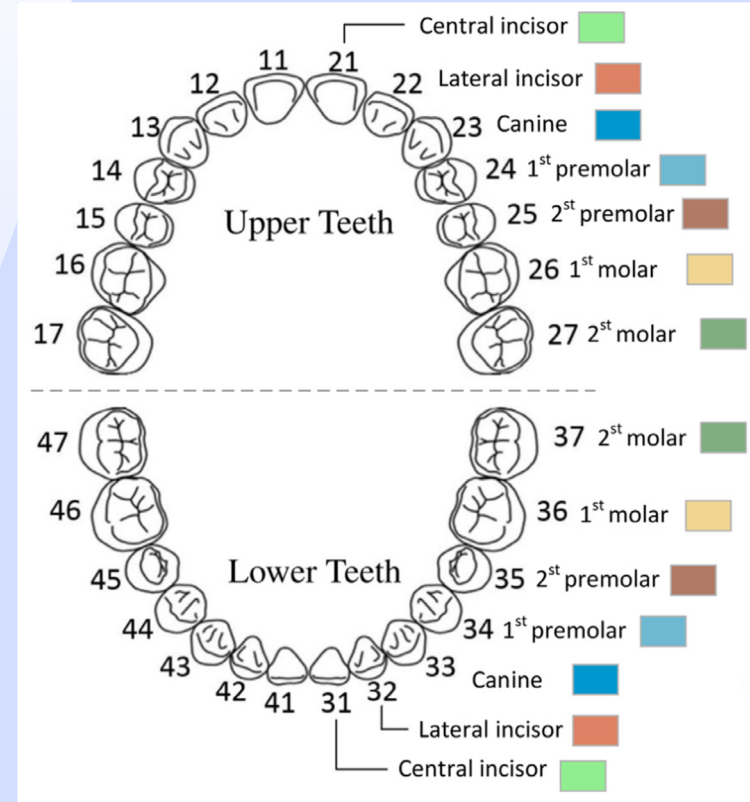


Fig. 1 ISO labeling of teeth^[1]

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Some Solution Attempts (General Pointcloud)

- PointNet^[2]
- PointNet++^[3]
- Dynamic Graph CNN^[4]
- Point Transformer^[5]

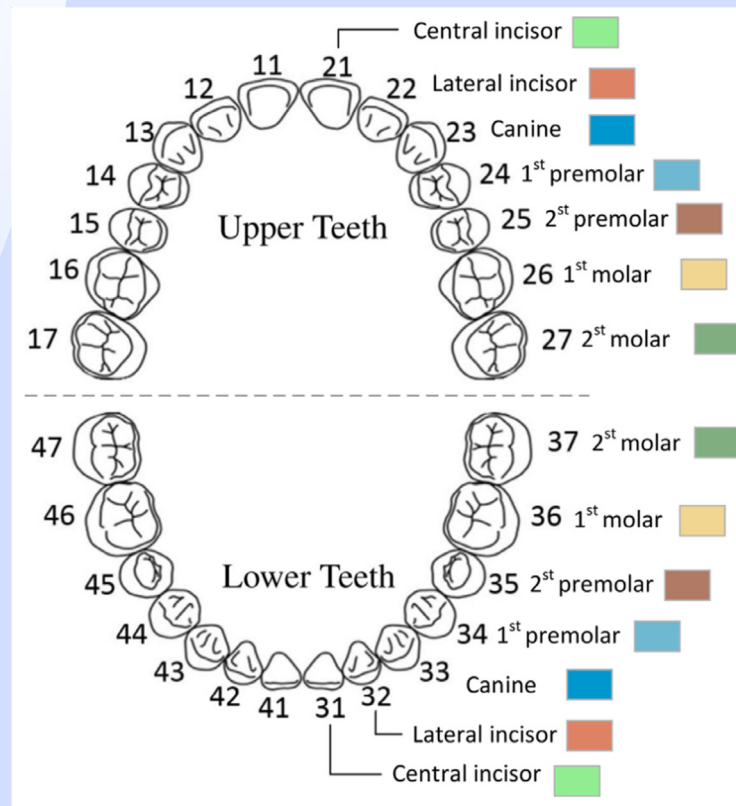


Fig. 1 ISO labeling of teeth^[1]

How about something specific for teeth?

Tooth Group Network: TSegNet^[1]

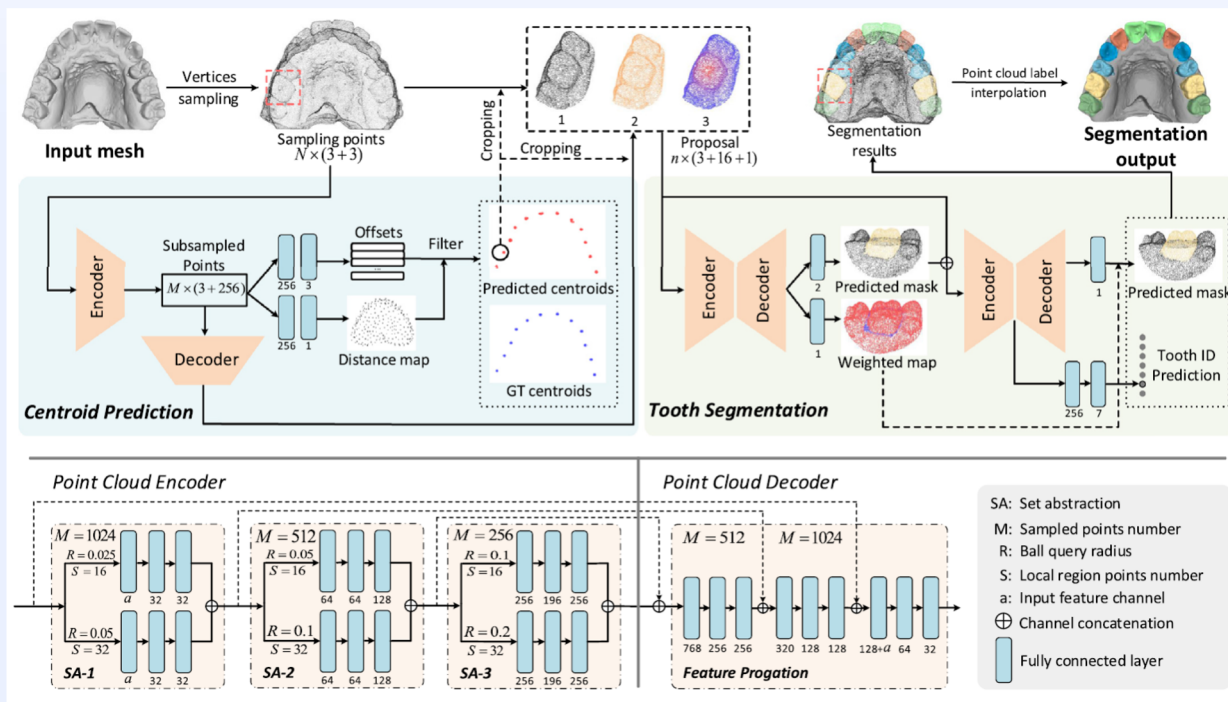
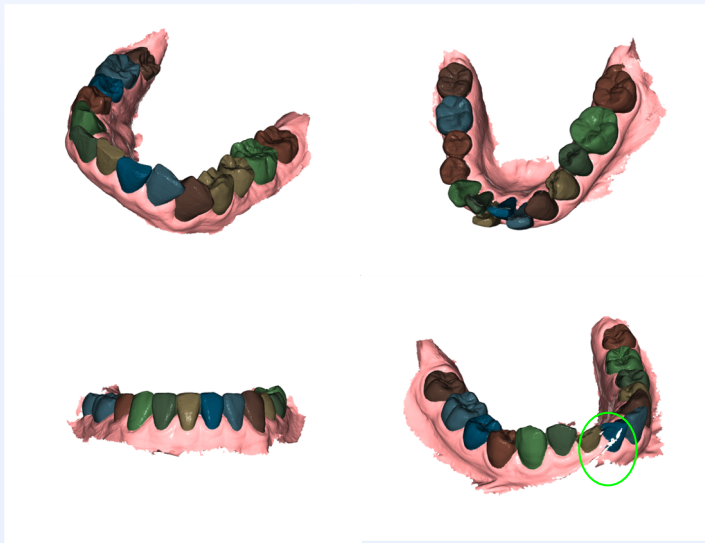


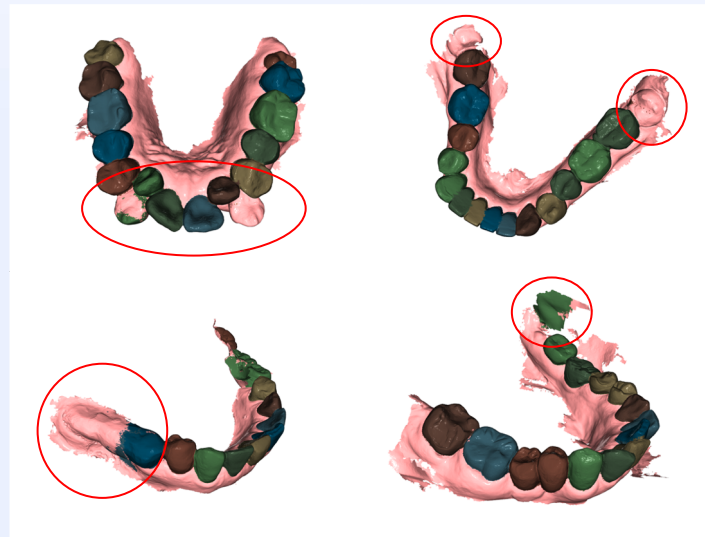
Fig. 3 TSegNet network diagram

Results of Pretrained Model^[7]

Good Results



Bad Results



Experience of Using TSegNet

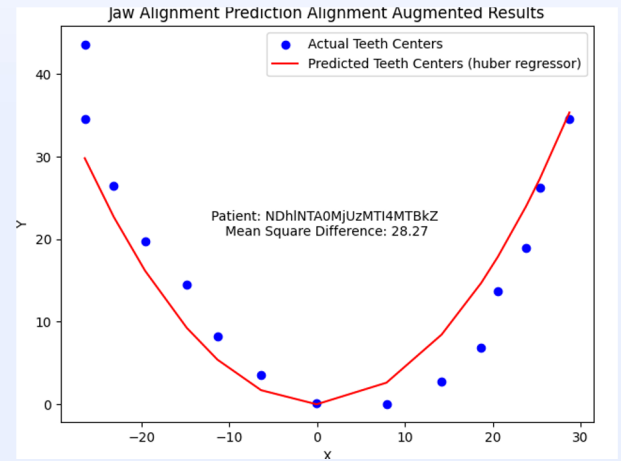
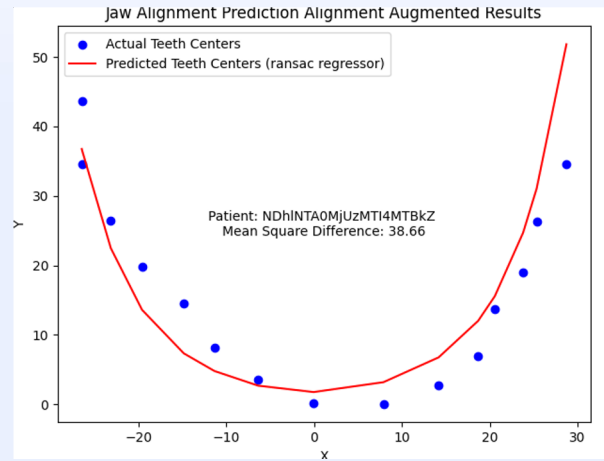
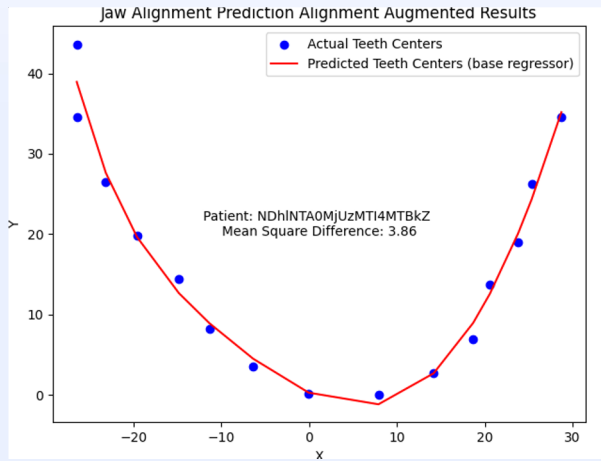
Pros

- Robust enough generally
- Easy to use
- Scriptable
- File formats that are easy to manipulate and extract data from
- See motivation slide

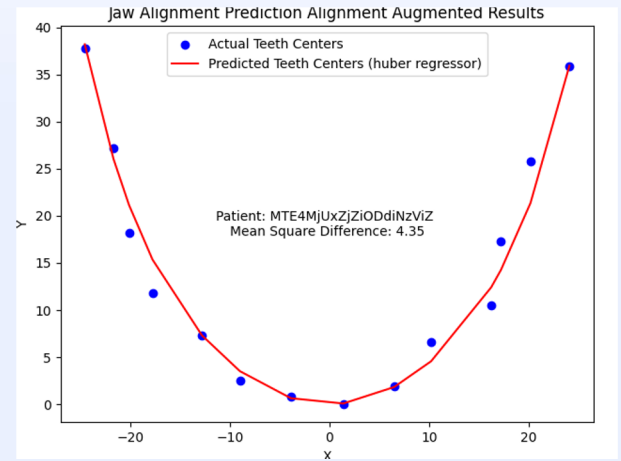
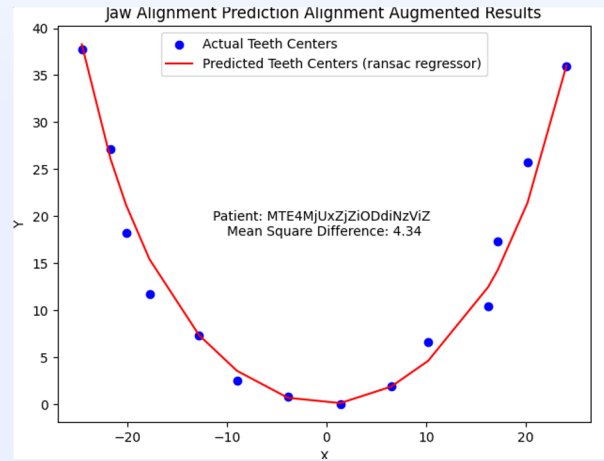
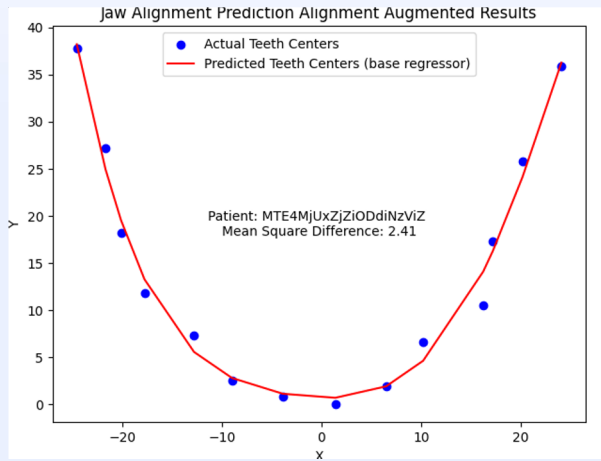
Cons

- Preprocessing required
 - Good scans required
 - Specific equipment required
 - Not accurate on weird teeth
-

Post Processing^[8]

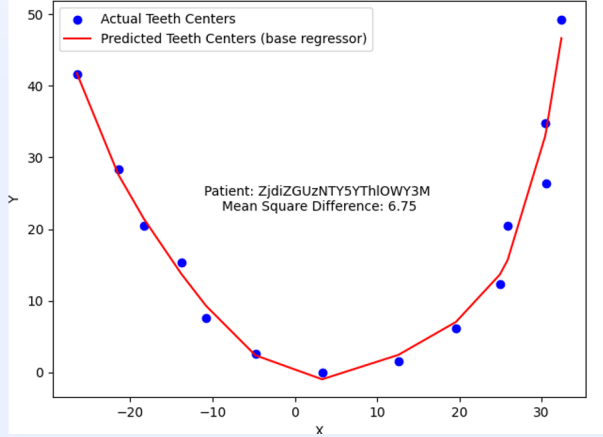


Post Processing^[8]

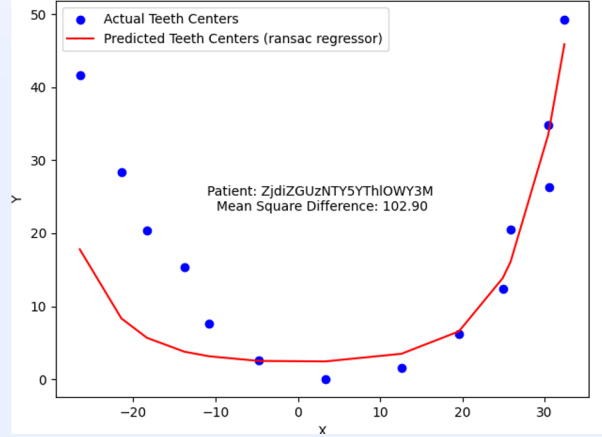


Post Processing^[8]

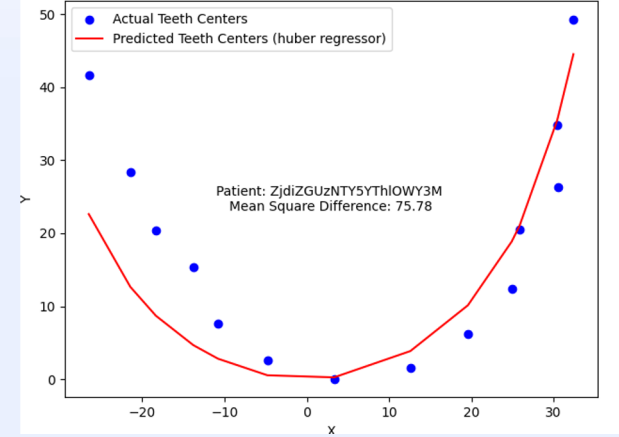
Jaw Alignment Prediction Alignment Augmented Results



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Jaw Alignment Prediction Alignment Augmented Results

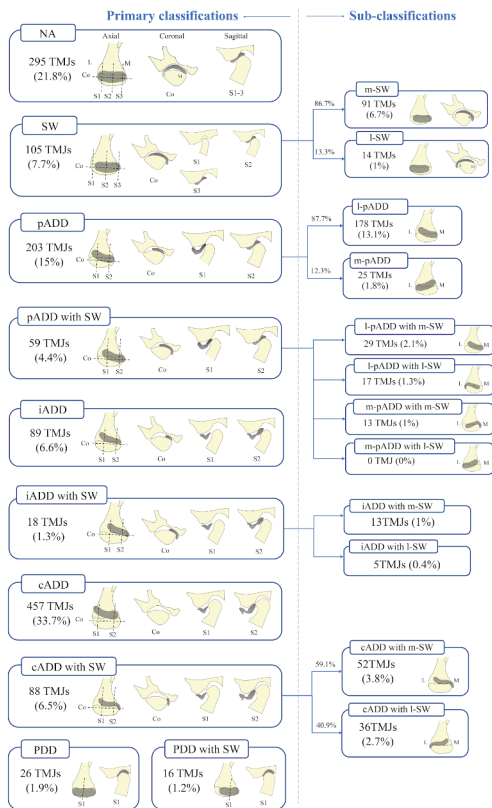




04.

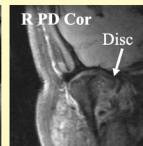
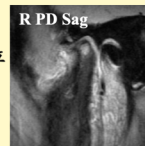
**MRI
Segmentation**

To Be Seen: What it Should Look Like**



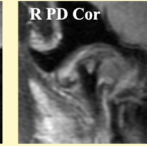
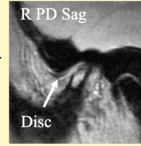
76歳, 女性
重度の内側転位

Osteophyte
SC or edema



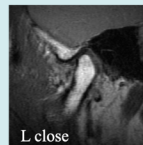
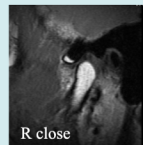
31歳, 女性
完全前方転位

Atrophy
SC or edema



73歳, 男性

No Abnormality



82歳, 男性

No Abnormality

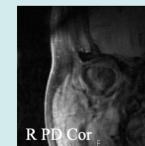
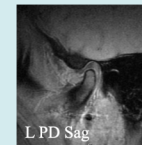
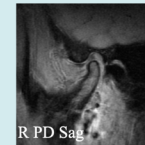
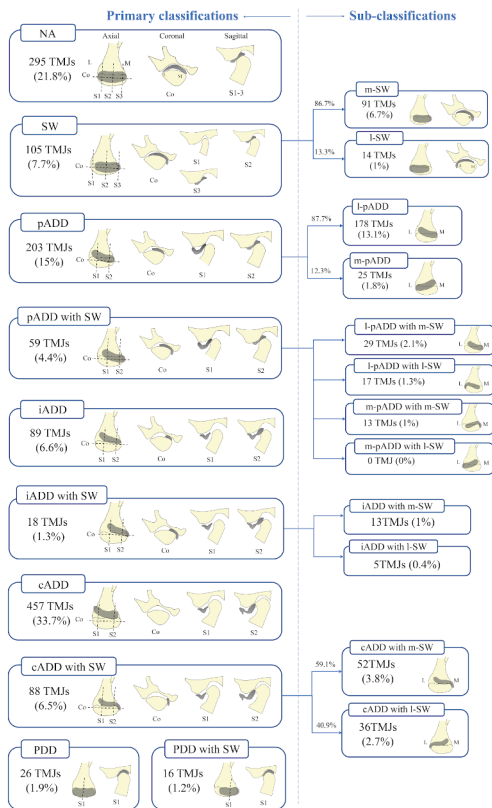


Fig. 4 Jaw joint classifications***

To Be Seen: What it Should Look Like**



Current Situation

- U-Net is the most popular with auxiliary methods of training and refining
- MRI segmentation is focused on brain imaging.

Fig. 4 Jaw joint classifications***

Thanks!

Do you have any questions?



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<https://github.com/dscpsyl>

